

LESSON 6-8 • SESSION 1 • PRACTICE • FORM A

Using Properties to Rewrite Expressions

Learning target – I can use the distributive property and like terms to rewrite an expression in simplest form. • 6.EE.A.3

REMEMBER

- **Distributive Property:** multiply the outside number by **every** term inside: $4(m + 2) = 4m + 8$.
- **Like terms** have the same variable part, so you can add them: $4m + 3m = 7m$. Add the plain numbers too.
- Swapping (Commutative) and regrouping (Associative) work for **adding and multiplying only**.

WORKED EXAMPLE

Simplify $4(m + 2) + 3m$.**Step 1** Distribute the 4: $4m + 8 + 3m$.**Step 2** Move like terms together: $4m + 3m + 8$.**Step 3** Combine: $7m + 8$. Test $m = 1$: $4(3) + 3 = 15$ and $7 + 8 = 15$.**Answer** $7m + 8$

WORDS **Distributive Property** (*propiedad distributiva*) • **like terms** (*términos semejantes*) • **equivalent expressions** (*expresiones equivalentes*)

PRACTICE

show your work • label units

1. Write each expression in simplest form.

a. $4(n + 5) =$ _____

b. $3(x + 2) + 5x =$ _____

c. $6(y + 3) + 9y =$ _____

d. $2(a + 4) + 3(a + 1) =$ _____

2. Name the property that allows each rewrite.

a. $8 + (x + 2) = (8 + x) + 2$ _____

b. $4 \times y = y \times 4$ _____

c. $y \times 1 = y$ _____

- 3.
- Team Cost.**
- A coach writes a team's cost as
- $3(n + 4) + 5n$
- , where
- n
- is the number of players.

Hint Multiply 3 by n and by 4.

a. Distribute: $3(n + 4) =$ _____

b. Combine like terms. Simplest form: _____

c. Check with $n = 2$. Original ____ New form ____

Show your work.

- 4.
- Simplify Fully.**
- Simplify
- $5(p + 3) + 2p + 8$
- .

Hint Add 15 and 8 after you distribute.

Show your work.

Simplest form _____

- 5.
- Simplify Fully.**
- Simplify
- $7(q + 2) + 3q + 5$
- .

Show your work.

Simplest form _____

- 6. Amira's Claim.** Amira says $24 \div 6$ and $6 \div 24$ are equivalent because of the Commutative Property. Find both values. Is she right?

Hint Does the Commutative Property cover division?

- 7. Three Rewrites.** Start with $3(w + 2) + w$. Write three different equivalent expressions. Name the property used for each.

Hint Try distributing, then reordering, then combining.

Show your work.

- 8. Two Tickets.** A family orders $3(a + 6)$ tickets and then $2(a + 4)$ tickets, where a is the number of adult tickets in each order.

- a.** Simplify each expression. _____
- b.** Write and simplify the total. _____
- c.** Find the total when $a = 5$. _____ tickets

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Tessa's Work.** Tessa rewrote $4(q + 5)$ as $4q + 5$. She said she multiplied the first term inside the parentheses. Explain her mistake. Then rewrite it correctly.

TEST PRACTICE

choose the best answer

- 10.** Which is equal to $5(x + 4)$?

A $5x + 4$ **B** $5x + 20$ **C** $x + 20$ **D** $9x$

- 11.** Which is $3(n + 2) + 4n$ in simplest form?

A $7n + 2$ **B** $7n + 6$ **C** $3n + 6$ **D** $13n$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

15y + 18 P	8n + 12 R	10q + 7 Z	5a + 26 T	28 O	8n + 4 Q	7p + 23 P	51 Y	10q + 19 E	1/4 R
#1c	#3b	#3c	#4	#5	#6	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____